# A Cross-Modality Feature Adaptive Interaction Approach for RGB-Infrared Object Detection in Aerial Imagery

Chushi Yu, *Graduate Student Member, IEEE*, and Yoan Shin, *Senior Member, IEEE*

***Abstract*—Object detection in aerial imagery, particularly from unmanned aerial vehicles (UAVs) and remote sensing platforms, is crucial but faces significant challenges such as modality misalignment, feature fusion degradation, and high computational complexity. To address these issues, this article introduces cross-modality feature adaptive detection (CMFADet), a novel framework for robust RGB-infrared (IR) object detection across diverse aerial scenarios. CMFADet improves feature learning through its innovative spatial–frequency feature enhancement module (SFEM) and IR adaptive feature aggregation block (IR-AFAB). It also integrates a channel interaction fusion (CIF) module for dynamic weight allocation, ensuring truly complementary information integration and avoiding mutual interference. This allocation is governed by the specific characteristics of the target and the inherent strengths of each modality. Detection accuracy is further refined via an adaptive task-aware alignment head (ATAH) that learns the joint features. Extensive experiments on the DroneVehicle, vehicle detection in aerial imagery (VEDAI), and OGSOD-1.0 datasets demonstrate CMFADet's superior performance, consistently surpassing state-of-the-art algorithms, and effectively addressing the aforementioned challenges. The source code for this work is publicly available at https://github.com/Yooyoo95/CMFADet**

***Index Terms*—Channel interaction fusion (CIF), cross-modality, feature extraction, RGB-infrared (IR) object detection.**

## I. Introduction

OBJECT detection is a foundational task in computer vision, underpinning a broad spectrum of real-world applications such as autonomous navigation, robotic perception, surveillance systems, and medical image interpretation [1], [2]. The advent of aerial platforms, including unmanned aerial vehicles (UAVs) and satellites, has revolutionized data acquisition and analysis across diverse domains, offering an unparalleled ability to rapidly identify and track targets. Object detection from these aerial perspectives is pivotal for critical tasks ranging from military defense, natural disaster relief, urban surveillance, and intelligent transportation systems.

However, aerial object detection, whether from low-altitude UAVs or high-altitude remote sensing platforms, presents inherent and significant challenges. These include the pervasive issue of small-object sizes, where targets often occupy only a few pixels, leading to severe information loss during feature extraction [3]. Furthermore, dynamic backgrounds, motion blur (especially from UAVs), varying illumination conditions, occlusions, and background clutter collectively complicate accurate and robust detection [4], [5].

Traditional object detection algorithms, even with recent advances in deep learning (DL) such as faster region-based convolutional neural network (faster R-CNN) [6], you only look once (YOLO) [7], and detection transformer (DETR) [8], primarily rely on single-modality data (e.g., RGB images). While RGB images offer rich texture, color information, and fine spatial details under optimal lighting, their performance significantly deteriorates in low-light conditions, adverse weather, or when objects are heavily occluded. In contrast, infrared (IR) imaging captures thermal radiation emitted by targets, enabling effective operation in low visibility environments and through occlusions like fog or smoke, providing crucial thermal signatures. However, IR images typically lack the texture and color details necessary for distinguishing small targets and mitigating false alarms. This inherent information bottleneck of single-modality data highlights the necessity for more comprehensive solutions.

To overcome these limitations, multimodal object detection has been introduced as a key approach. By fusing data from different sensors like visible light, IR, and synthetic aperture radar (SAR), multimodal methods leverage the complementary information between modalities and improve detection accuracy and robustness in complex aerial environments. The DroneVehicle dataset is the first large-scale drone-based RGB-IR image dataset for vehicle detection in complex, full-time scenarios [9]. It provides valuable resources for multimodal research and serves as an important benchmark. Similarly, vehicle detection in aerial imagery (VEDAI) dataset [10] is a widely recognized benchmark for small vehicle detection in the remote sensing field. Both DroneVehicle and VEDAI represent typical scenes from low-altitude drone perspectives and high-altitude remote sensing platforms, respectively. The optical image guided SAR image object

Received 10 November 2025; revised 3 January 2026; accepted 17 January 2026. Date of publication 23 January 2026; date of current version 11 February 2026. This work was supported in part by the National Research Foundation (NRF) of Korea funded by Korean Government [Ministry of Science and ICT (MSIT)] under Grant RS-2025-02214082; and in part by MSIT, South Korea, through the Information Technology Research Center (ITRC) Support Program supervised by the Institute for Information and Communications Technology Planning and Evaluation (IITP) under Grant IITP-2026-RS-2023-00258639. *(Corresponding author: Yoan Shin.)*

The authors are with the School of Electronic Engineering, Soongsil University, Seoul 06978, Korea (e-mail: csyu@soongsil.ac.kr; yashin@ssu.ac.kr).

Digital Object Identifier 10.1109/TGRS.2026.3657379

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

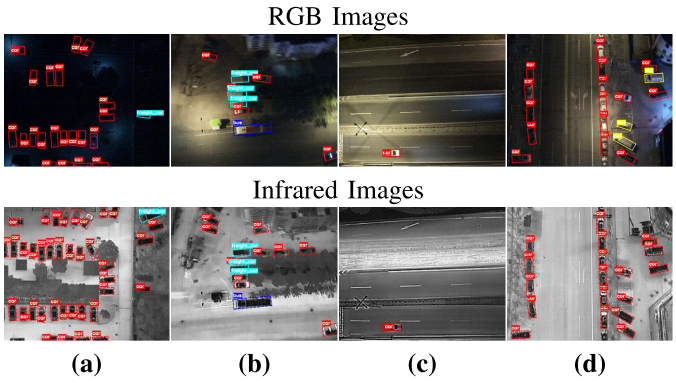



Fig. 1. Illustration of the limitations and challenges in multimodality object detection. The solid line bounding boxes represent RGB labels, and the dotted lines are IR labels on the DroneVehicle dataset. (a) Limitation of single mode in night environment. (b) Low visibility with occlusion. (c) Misalignment issue of fast moving targets. (d) Label category conflict of objects in images from different modalities.

detection (OGSOD-1.0) [11] is the first SAR–optical benchmark dataset for multiobject detection.

Despite promising advancements in multimodality fusion, several challenges persist. Fig. 1 illustrates these challenges using examples from the DroneVehicle dataset. First, due to differences in perspective, distance, and sensor characteristics, a weak misalignment challenge often exists between RGB and IR images, particularly for small targets. This misalignment requires careful consideration during fusion. Second, conventional multimodal fusion methods, primarily operating in the spatial domain, often miss valuable frequency-domain cues that could provide more robust representations, neglecting the distinct frequency characteristics of RGB and IR modalities. Additionally, existing object detector heads commonly employ independent classification and localization branches, which can lead to a lack of interaction and suboptimal joint feature learning between these two crucial tasks. Finally, the pursuit of performance often leads to computationally heavy models, hindering real-time deployment on aerial platforms.

To address the aforementioned challenges and significantly boost detection accuracy across diverse aerial scenarios, this article proposes an innovative object detection framework for RGB-IR aerial images, named cross-modality feature adaptive detection (CMFADet). At its core, CMFADet leverages spatial- and frequency-domain decomposition through a dedicated spatial–frequency feature enhancement module (SFEM) to isolate global structural information from fine-grained local details, while an IR adaptive feature aggregation block (IR-AFAB) is specifically designed to enhance feature extraction for IR imagery. Instead of relying on simple feature-level summation, CMFADet employs more sophisticated strategies for dynamic weight allocation, expertly managed by a channel interaction fusion (CIF) module. Furthermore, it tackles the limitations of independent classification and localization branches by proposing an adaptive task-aware alignment head (ATAH), which learns to integrate task-specific features from various convolutional layers. The main contributions are summarized as follows.

1) We propose the CMFADet framework for multimodality object detection in aerial images with a two-stream backbone, which incorporates an SFEM to enhance RGB feature fidelity against variance and an IR-AFAB to prevent thermal feature degradation and reinforce sparse IR features.
2) We introduce a CIF module to overcome static feature merging limitations. CIF facilitates adaptive aggregation by dynamically allocating fusion weights and promotes deep residual cross-modality feature exchange to mitigate mutual interference.
3) We design the ATAH as a unique detection head that performs adaptive task-aware alignment on task-specific features by implicitly modeling inter-task coupling to maximize predictive performance.
4) We conduct experiments on three representative multimodality object detection datasets, DroneVehicle, VEDAI, and OGSOD-1.0, demonstrating state-of-the-art performance with favorable computational efficiency, validating its suitability for aerial deployment.

The remainder of this article is organized as follows. Section II reviews related work on relevant DL object detection techniques and multimodal aerial object detection. In Section III, we describe the proposed architecture and mechanisms in detail. Section IV presents experimental results and analysis, including comparisons with state-of-the-art methods. Finally, we conclude this article in Section V.

## II. Related Work

### A. Generic Object Detection

Object detection is a core computer vision task focused on classifying and precisely localizing objects with bounding boxes. DL-based methods promote the development of this task, which can be broadly categorized into two- and single-stage detectors. Two-stage detectors such as the R-CNN family [6], [12], [13] first generate candidate regions and then classify and refine them, which achieves high accuracy but incurs substantial computational cost. In contrast, single-stage detectors, including single-shot multibox detector (SSD) [14], RetinaNet [15], and the YOLO series [16], [17], [18], directly predict object categories and locations in a single pass, providing superior speed. Improvements in backbone networks (e.g., residual neural network (ResNet) [19]), feature aggregation mechanisms in neck networks (e.g., feature pyramid network (FPN) [20] and path aggregation network (PANet) [21]), refined detection heads (e.g., dynamic head [22]), and optimized loss functions further enhance performance. Beyond these refinements, methods like deformable convolutional network (DCNv2) [23] enhance adaptability to object scale and shape variations, while task-aligned detectors like the task-aligned one-stage object detection (TOOD) [24] improve consistency between classification and localization by designing a task-aligned head and task alignment learning strategy. Building upon the real-time DETR (RT-DETR) [25], the MCG-RTDETR architecture enhances precise object detection in UAV imagery by integrating dual and deformable convolution modules, a cascaded group attention module,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

and a context-guided feature fusion block [26]. Despite these advances, existing generic object detectors mainly focus on architectural optimization and task alignment between classification and localization, while cross-modality discrepancy and modality-specific feature degradation are not explicitly addressed.

### B. Oriented Object Detection

Unlike generic object detection using horizontal bounding boxes (HBBs), oriented object detection employs oriented bounding boxes (OBBs) or quadrilaterals for precise localization of targets with arbitrary orientations. OBB overcomes limitations of HBB-based methods in dense scenes, complex backgrounds, or with high-aspect-ratio targets, especially in aerial and remote sensing imagery. The field has evolved from two-stage to single-stage architectures. RoI Transformer [27] pioneers spatial transformations and consists of the rotated region of interest (RRoI) Learner and RRoI Warping. Oriented R-CNN [28] later proposes an oriented region proposal network to generate high-quality oriented proposals. However, the computational cost of two-stage methods motivated the development of single-stage detectors. A single-shot alignment network ($S^2$A-Net) tackles feature misalignment with a feature alignment module (FAM) and an oriented detection module (ODM) [29]. ReDet [30] incorporates rotation-equivariant networks into the detector to extract rotation-equivariant features. Song et al. [31] introduce a rotation-invariant transformer plus model (RIFormer+) by designing a rotation-invariant cross-patch embedding (RICPE) module and a response-enhanced attention (REA) to enhance local and global feature learning. These developments highlight the importance of rotation modeling for oriented targets and provide valuable insights into how orientation-aware representations can facilitate effective detection under complex imaging conditions.

### C. Multimodal Object Detection

Multimodal object detection integrates complementary cues from heterogeneous sensors (e.g., visible, IR, and depth) to improve robustness where single modalities are insufficient. The key challenge lies in effectively fusing diverse data and mitigating cross-modal discrepancies. Fusion strategies are categorized by processing stages. Some fusion methods combine raw data channels directly by concatenating multimodality images as a preprocessing process. In decision level, each modality is detected independently, and predictions are combined (e.g., weighted non-maximum suppression). Feature-level fusion is the most common approach, balancing interaction and modality-specific processing by integrating extracted features (e.g., via concatenation or attention mechanism [32]). In feature-level fusion, detectors prioritize alignment and conflict resolution. UA-CMDet [9] introduces an uncertainty-aware mechanism to evaluate the uncertainty weights of every target for RGB and IR modalities by calculating the cross-modal intersection over union (IoU) and the illumination value. The two-stream feature alignment detector (TSFADet) addresses misalignment challenge by calibrating feature maps via a translation-scale-rotation alignment (TSRA) module [33]. In [34], a transformer-based calibrated and complementary method named $C^2$Former can simultaneously alleviate the modality miscalibration and fusion inaccuracy. The Dual-YOLO [35] is an IR object detector based on YOLOv7, which introduces the inception and SE modules into the attention fusion method to strengthen the characteristic complementarity. The cross-modal conflict-aware learning network (CALNet) [36] is a multispectral oriented object detector designed to tackle the semantic conflict problem. For illumination changes, Shang et al. [37] propose a cross-modality and cross-domain low-light detector (CCLDet) consisting of an illumination-aware module (OIAM), a visibility loss and cross-domain feature fusion module (CDFFM). Nie et al. [38] present a robust CNN–Transformer-based framework for cross-modal feature fusion in visual and IR imagery, focusing on the effective interaction between local and long-range features. Overall, existing multimodal object detectors typically rely on explicitly designed fusion or alignment modules to handle heterogeneous discrepancy, with cross-modal interactions governed by the predefined model structure.

### D. Related Multimodal Applications and Insights

Related multimodal applications in computer vision and remote sensing offer complementary perspectives on handling heterogeneous modalities, with recent advances in tasks such as salient object detection (SOD) providing relevant references for multimodal detection. The CONTRINET [39] utilizes a divide-and-conquer strategy with a triple-flow network to handle modality-specific cues and enhance resilience to defective data. The transformer-based CATNet [40] and CPNet [41] tackle multiscale aggregation challenges. Specifically, CATNet uses a cascaded and aggregated decoder to mitigate information loss, while CPNet employs a progressive decoding strategy to refine predictions. Beyond SOD, GCViTDet [42] demonstrates that generalized vision transformers with masked token modeling can enhance semantic fusion.

General multimodal remote sensing research focuses on robust low-level feature handling. The content-adaptive nonlocal convolution (CANConv) [43] shows the utility of spatially adaptive convolution and nonlocal self-similarity for effective image fusion. The co-occurrence filter-based matching work [44] confirms the importance of building feature representations robust to nonlinear radiation differences (NRDs) for accurate multimodal image registration. The anchor-guided fuzzy clustering study [45] shows the effectiveness of learning highly discriminative, low-dimensional multimodal features to enable robust high-level interpretation. In hyperspectral scene classification, the HRSS-C [46] employs a three-branch spectral–spatial fusion network to leverage diverse feature representations. These findings further motivate considering adaptive alignment and modality-specific feature modeling. From an application perspective, aerial person detection (APD) represents another relevant application domain. A recent survey [47] categorizes APD methods into object-aware, sample-oriented, fusion-based, and lightweight designs. For instance, a lightweight distillation network (APDNet) [48] validates an effective approach to balance detection speed and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

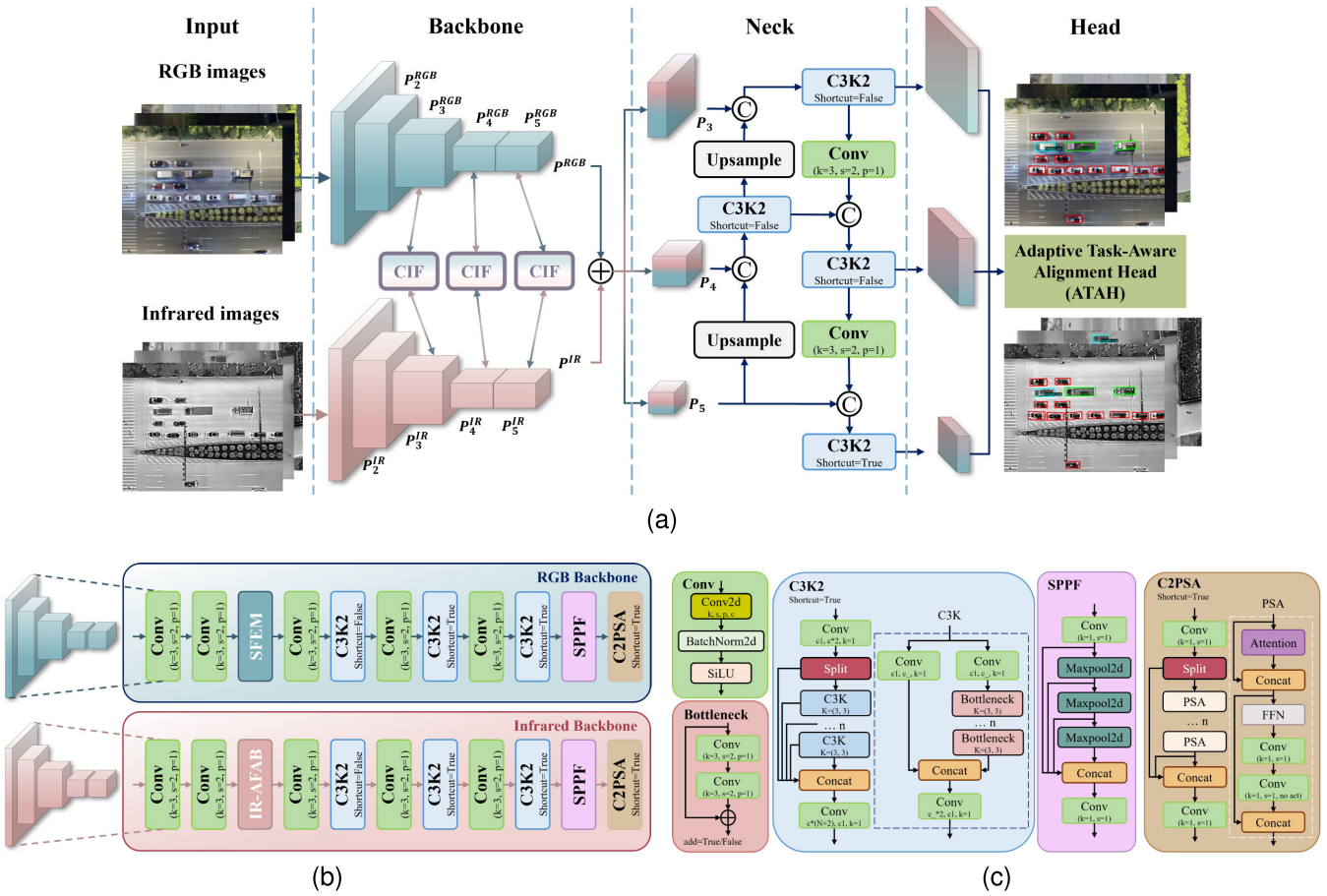



Fig. 2. (a) Overall architecture of the proposed method CMFADet. (b) RGB and IR backbone network in detail. (c) Structure of the Conv, C3K2, SPPF, and C2PSA blocks.

accuracy for edge deployment through the teach-assistant distillation (TAD) and low-precision quantization (LPQ) modules. VTSaRNet [49], designed for visible–thermal search-and-rescue scenarios, employs a union transformer network to fuse global features from visible and thermal modalities, improving detection of sparsely distributed person instances. These related multimodal applications demonstrate that effective handling of heterogeneous modalities has been explored across different vision tasks, providing useful references for multimodal object detection.

## III. Proposed Method

This section introduces the CMFADet approach engineered for enhanced RGB-IR object detection within drone and remote sensing imagery. This is achieved by integrating synergistic components into a cohesive end-to-end framework. As depicted in Fig. 2, the proposed architecture is composed of a backbone network, a neck network, and prediction heads. Initially, the multimodality backbone processes input RGB-IR paired images, extracting comprehensive information. This backbone then generates multiscale feature maps from three distinct stages ($P_3$–$P_5$) through the synergistic operation of the CIF module and element-wise addition. Subsequently, these enriched feature maps are propagated to the neck network, where they undergo sophisticated feature fusion. Finally, the ATAH is employed to perform precise object localization and classification as the prediction head.

### A. Multimodality Backbone

The backbone network comprises two parallel network structures, specifically designed to process RGB and IR images, respectively. Paired images are fed into their corresponding network branches. In the initial stage, we design dedicated modules to enhance feature extraction for the distinct characteristics of each image modality. Specifically, an SFEM is utilized in the RGB branch, while an IR-AFAB is proposed for the IR branch. Then, the method retains the cross-stage partial with kernel size 2 (C3K2) block consisting of two cross stage partial (CSP) bottlenecks, the spatial pyramid pooling-fast (SPPF) block, and the convolutional block with parallel spatial attention (C2PSA) block from the original YOLOv11 version. Detailed explanations of these modules are provided in the following.

*1) Spatial–Frequency Feature Enhancement Module:* The SFEM is designed to improve feature representations by synergistically integrating spatial- and frequency-domain information through two parallel branches as shown in Fig. 3. This decomposition mechanism is critical for RGB modality, as the frequency branch introduces frequency-domain feature transformations that facilitate global structural rep-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

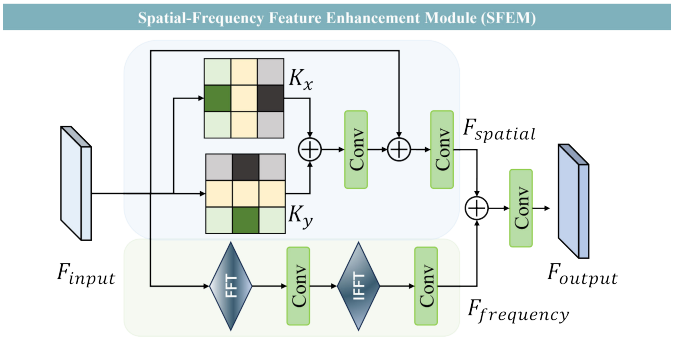



Fig. 3. Architecture of the SFEM.

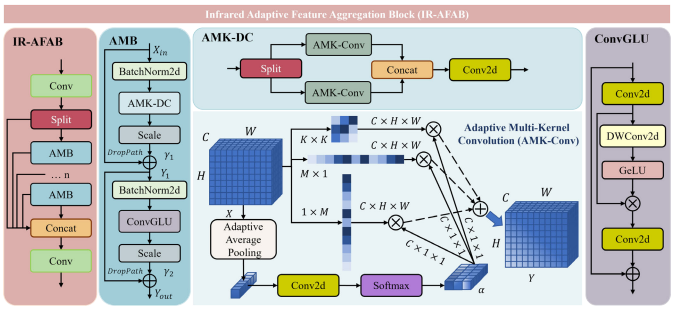



Fig. 4. Architecture of the IR-AFAB.

resentation, while the spatial branch extracts essential local boundary details. This separation ensures feature robustness under diverse aerial conditions.

In the spatial branch, the ScharrConv operation is utilized for horizontal ($G_x$) and vertical ($G_y$) gradient computation using predefined Scharr kernels [50]. The horizontal ($K_x$) and vertical ($K_y$) Scharr kernels are defined as follows:

$$K_x = \begin{bmatrix} +3 & 0 & -3 \\ +10 & 0 & -10 \\ +3 & 0 & -3 \end{bmatrix}, \quad K_y = \begin{bmatrix} +3 & +10 & +3 \\ 0 & 0 & 0 \\ -3 & -10 & -3 \end{bmatrix}. \tag{1}$$

For an input feature map $F_{\text{input}}$, the operation first computes the horizontal gradient ($G_x = F_{\text{input}} * K_x$) and vertical ($G_y = F_{\text{input}} * K_y$) gradient, where $*$ denotes the convolution operation. The combined gradients ($G_x + G_y$) are then processed through two convolutional layers. The resulting feature map is subsequently added to the original input $F_{\text{input}}$ via a residual connection to preserve low-level information, yielding the final spatial feature $F_{\text{spatial}}$.

In the frequency branch, the input $F_{\text{input}}$ is first transformed into the frequency domain via a 2-D fast Fourier transform ($\text{FFT}_{\text{2-D}}$). The transformed frequency features are then fed into a convolutional layer, which enables learnable feature transformation in the frequency domain. The processed frequency features are then transformed back to the spatial domain via a 2-D inverse fast Fourier transform ($\text{IFFT}_{\text{2-D}}$) and subsequently passed through an additional convolutional layer. Finally, the output of the spatial branch ($F_{\text{spatial}}$) and the frequency branch ($F_{\text{frequency}}$) are combined via an element-wise addition and then passed through a final convolutional layer to integrate local edge and texture information with global frequency patterns, producing the module's final output $F_{\text{output}}$. The process can be described as follows:

$$F_{\text{spatial}} = \text{Conv}\left(\text{Conv}\left(G_x + G_y\right) + F_{\text{input}}\right) \tag{2}$$

$$F_{\text{frequency}} = \text{Conv}\left(\text{IFFT}_{\text{2-D}}\left(\text{Conv}\left(\text{FFT}_{\text{2-D}}\left(F_{\text{input}}\right)\right)\right)\right) \tag{3}$$

$$F_{\text{output}} = \text{Conv}\left(F_{\text{spatial}} + F_{\text{frequency}}\right). \tag{4}$$

*2) IR Adaptive Feature Aggregation Block:* We propose the IR-AFAB to enhance the feature extraction for IR imagery. As shown in Fig. 4, the IR-AFAB operates by splitting the input into two pathways: a deep branch and a skip connection. The deep branch is processed by a sequence of cascaded adaptive mixer blocks (AMBs), while the skip connection is implemented to prevent thermal feature degradation in the deep path. These pathways are fused via concatenation followed by a $1 \times 1$ convolution. This design effectively integrates features from both shallow and deep processing streams to preserve sparse, low-level thermal features.

Inspired by MetaFormer [51], the AMB employs a sequential combination of spatial and channel mixing pathways with residual connections. To stabilize the training and regularize the model, we incorporate a scale mechanism and a DropPath (DP) operation in each residual branch [52], [53]. The AMB processes features in two consecutive stages. The first stage, using an adaptive multi-kernel depthwise convolution (AMK-DC), aggregates spatial information. The second stage, using a convolutional gated linear unit (ConvGLU), enhances channel discriminability. The output of each stage is added to its input through a residual connection. This sequential structure allows the AMB to first aggregate spatial information and then enhance channel discriminability, forming a powerful feature representation. The process can be formulated as follows:

$$Y_1 = X_{\text{in}} + \text{DP}\left(\gamma_1 \odot \text{AMK-dc}\left(\text{BN}\left(X_{\text{in}}\right)\right)\right) \tag{5}$$

$$Y_{\text{out}} = Y_1 + \text{DP}\left(\gamma_2 \odot \text{ConvGLU}\left(\text{BN}\left(Y_1\right)\right)\right) \tag{6}$$

where $X_{\text{in}}$ is the input feature to the AMB module, BN is the batch normalization layer, and $\gamma_1$ and $\gamma_2$ are learnable layer-scale parameters. DP($\cdot$) denotes the DP operation. AMK-dc($\cdot$) and ConvGLU($\cdot$) represent the spatial and channel mixing stages, respectively.

The AMK-DC module splits the input channels into distinct groups. Each group is processed by an adaptive multi-kernel convolution (AMK-Conv) with varying kernel sizes (e.g., $3 \times 3$ and $5 \times 5$). The outputs from these parallel streams are concatenated and fused through a $1 \times 1$ pointwise convolution. This design captures multiscale and multidirectional spatial features. The AMK-Conv generates dynamic weights for three heterogeneous kernels (square, horizontal, and vertical) based on the input's global context. It uses adaptive average pooling and a lightweight $1 \times 1$ convolution to produce weighting coefficients. These coefficients are normalized by a Softmax function and applied element-wise to each depthwise convolution branch. The adaptive multikernel design is essential to reinforce sparse IR features by ensuring multiscale receptive fields and dynamically prioritizing kernels most responsive to small thermal targets. The final outputs are then summed. The process can be formulated as follows:

$$\alpha = \text{Softmax}\left(\text{Conv}_{1\times1}\left(\text{AAP}\left(X\right)\right)\right) \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

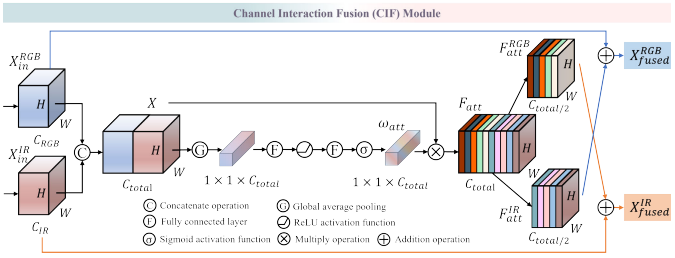



Fig. 5. Architecture of the proposed CIF module.

$$Y = \sigma\left(\mathrm{BN}\left(\sum_{i=1}^{3} \text{AMK-Conv}_i\,(X) \odot \alpha_i\right)\right) \tag{8}$$

where $X \in \mathbb{R}^{C\times H\times W}$ is the input feature map and AAP$(\cdot)$ and $\text{Conv}_{1\times1}(\cdot)$ are used to extract global context for dynamic weight generation. The index $i = 1, 2, 3$ corresponds to the square, horizontal, and vertical depthwise convolution branches, respectively. $\odot$ denotes the element-wise multiplication and $\sigma(\cdot)$ is a sigmoid linear unit (SiLU) activation.

The ConvGLU module, similar to that in TransNeXt [54], enhances inter-channel interactions. It uses a gating mechanism where two convolutional layers process the input, one acts as a gate and the other extracts features. Their element-wise multiplication yields a weighted feature tensor, which is added to the input via a residual connection. This mechanism improves feature discriminative power, which is crucial for information in IR images.

## B. CIF Module

To effectively integrate the complementary information from the RGB and IR modalities, we propose a CIF module. This module is designed to enhance feature representation by learning inter-channel dependencies across modalities and facilitating a deeper exchange of crucial information. By generating channel-wise weights from global contextual information, the CIF module achieves dynamic adaptive fusion, a mechanism necessary for robust performance in heterogeneous aerial scenes.

As illustrated in Fig. 5, the CIF module takes the RGB feature map $X_{\text{in}}^{\text{RGB}} \in \mathbb{R}^{B\times C_{\text{RGB}}\times H\times W}$ and the IR feature map $X_{\text{in}}^{\text{IR}} \in \mathbb{R}^{B\times C_{\text{IR}}\times H\times W}$ as input, where $B$, $H$, and $W$ represent the batch size, height, and width of the feature maps, respectively. These are first concatenated along the channel dimension to form a combined feature map $X \in \mathbb{R}^{B\times C_{\text{total}}\times H\times W}$, where $C_{\text{total}} = C_{\text{RGB}} + C_{\text{IR}}$. First, a global average pooling (GAP) operation is applied on the input feature map $X$ to capture global contextual information for each channel. This operation reduces the spatial dimensions ($H \times W$) to $1 \times 1$. Next, these pooled features are fed into an excitation block to generate channel-wise weights $\omega_{\text{att}} \in \mathbb{R}^{B\times C_{\text{total}}\times 1\times 1}$, which consists of two fully connected layers with activation layers (a ReLU and a sigmoid). Then, the attention-weighted feature $F_{\text{att}}$ is calculated by the weight $\omega_{\text{att}}$ and the original input feature map $X$ through an element-wise multiplication $\odot$. The cross-modality interaction mechanism with residual connection is the key innovation of the CIF module. Instead of simply feedback $F_{\text{att}}$, we facilitate a deeper communication of information between RGB and IR branches. We split the attention-weighted feature into $F_{\text{att}}^{\text{RGB}}$ and $F_{\text{att}}^{\text{IR}}$ and recombine these with input RGB and IR feature maps through a residual connection following the cross-modality mechanism. The outputs of the CIF module are described as follows.

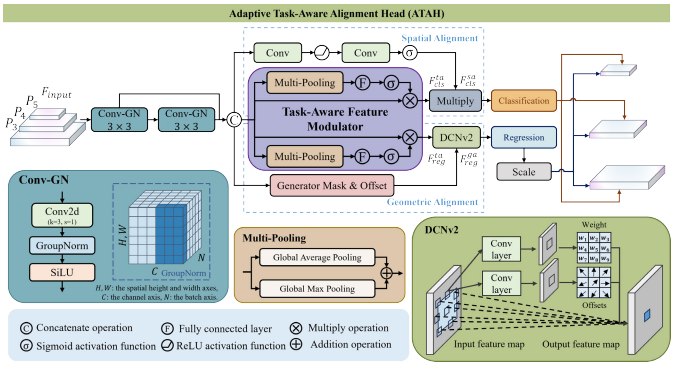



Fig. 6. Structure of the proposed ATAH.

$$X_{\text{fused}}^{\text{RGB}} = X_{\text{in}}^{\text{RGB}} + F_{\text{att}}^{\text{IR}} \tag{9}$$

$$X_{\text{fused}}^{\text{IR}} = X_{\text{in}}^{\text{IR}} + F_{\text{att}}^{\text{RGB}} \tag{10}$$

where $F_{\text{att}} = X \odot \omega_{\text{att}}$ and $(F_{\text{att}}^{\text{RGB}}, F_{\text{att}}^{\text{IR}}) = \text{Split}(F_{\text{att}})$.

By integrating channel attention with this unique cross-modality residual interaction, the CIF module not only ensures that the original feature information is preserved but also mitigates the detrimental effect of feature misalignment while facilitating smoother gradient flow during training. This adaptive fusion strategy provides a robust solution for effective cross-modality information exchange.

## C. Adaptive Task-Aware Alignment Head

To address the inherent misalignment between generic feature representations and task-specific requirements, we propose the ATAH. This head module enhances detection performance by combining implicit inter-task feature coupling with dynamic, task-specific feature alignment. The ATAH's overall structure is depicted in Fig. 6.

Input feature maps, $F_{\text{input}}$, from different levels of the neck network are first processed through a block of two convolutional layers with group normalization (Conv-GN). This shared block contributes to the parameter-efficient nature of our CMFADet by promoting weight reuse across different scales and, critically, establishes a feature base for inter-task coupling through joint optimization. Subsequently, two parallel branches are constructed using two identical task-aware feature modulator (TAFM) modules to generate task-specific features. The multi-pooling component provides rich multiscale contextual information by obtaining GAP and global max pooling (GMP) features and then adding the results. These modules output two distinct representations: $F_{\text{cls}}^{\text{ta}}$ for classification and $F_{\text{reg}}^{\text{ta}}$ for localization. This implicit feature coupling, driven by the shared Conv-GN weights and a unified multi-task loss, is the source of the inter-task interaction. Each branch then performs a specialized alignment operation. The classification branch employs a spatial alignment block. A spatial attention map is dynamically produced from the unified features through

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

a lightweight convolutional subnetwork. The map modulates $F_{\text{cls}}^{\text{ta}}$ through element-wise multiplication, yielding a spatially aligned representation, $F_{\text{cls}}^{\text{sa}}$, with enhanced discriminability and focus by selectively amplifying features in key regions. Meanwhile, the regression branch utilizes a geometric alignment block. A DCNv2 is applied to $F_{\text{reg}}^{\text{ta}}$, with spatial sampling offsets and modulation masks dynamically predicted from the input feature maps. This produces a geometrically aligned representation, $F_{\text{reg}}^{\text{ga}}$, whose receptive field adaptively conforms to the irregular geometry and orientation of objects. Furthermore, this adaptive sampling mechanism allows DCNv2 to implicitly compensate for small spatial misalignment inherent in the fused RGB-IR features, serving a critical dual role in achieving both task-specific and inter-modality alignment.

The classification branch outputs category confidence scores through a convolutional layer followed by a sigmoid activation function. The regression branch predicts the bounding box as a 4-tuple $(x, y, w, h)$ using a distribution-based representation, with coordinates decoded via the distribution focal loss (DFL) [55]. For OBB, an additional regression head predicts a raw angle logit $x_\theta$, which is mapped to the angular range $[-\pi/4, 3\pi/4]$ to resolve rotational ambiguity and ensure consistency with object alignment, as defined in the following:

$$\theta = \left(\text{sigmoid}\left(x_\theta\right) - 0.25\right) \cdot \pi. \tag{11}$$

During inference, predictions from all pyramid levels are concatenated, and the final OBBs $(x, y, w, h, \theta)$ are constructed from the decoded outputs. The classification and regression prediction branches proceed in parallel to ensure task specificity and prevent mutual interference. However, the overall performance relies on the effective execution of task-aware alignment on the implicitly coupled features, driving superior joint prediction quality.

### D. Loss Function

The overall training objective is a weighted sum of three distinct components: a classification loss ($L_{\text{cls}}$), a regression loss ($L_{\text{obb}}$), and a DFL ($L_{\text{dfl}}$). The total loss $L$ is defined as follows:

$$L = \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{obb}} \cdot L_{\text{obb}} + \lambda_{\text{dfl}} \cdot L_{\text{dfl}} \tag{12}$$

where $\lambda_{\text{cls}}$, $\lambda_{\text{obb}}$, and $\lambda_{\text{dfl}}$ are hyperparameters that control the contribution of each loss component.

The classification loss is calculated using a binary cross-entropy (BCE) loss on the logits to correctly identify the object class for each predicted box

$$L_{\text{cls}} = \frac{1}{N} \sum_i -\left(y_i \log\left(\hat{y}_i\right) + \left(1 - y_i\right) \log\left(1 - \hat{y}_i\right)\right) \tag{13}$$

where $y_i$ and $\hat{y}_i$ are the ground truth and predicted class probabilities, respectively, and $N$ is the number of positive samples.

For regression, we use the probabilistic IoU (ProbIoU) loss, which is specifically designed for rotated bounding boxes by treating each bounding box as a 2-D Gaussian distribution [56]

$$B_C\left(p, q\right) = \int_{\mathbb{R}^2} \sqrt{p\left(x\right) q\left(x\right)} dx \tag{14}$$

$$H_D\left(p, q\right) = \sqrt{1 - B_c\left(p, q\right)} \tag{15}$$

$$L_{\text{obb}} = H_D\left(p, q\right) = 1 - \text{ProbIoU}\left(p, q\right) \in [0, 1] \tag{16}$$

where $p(x)$ and $q(x)$ are the probability density functions of predicted and ground truth bounding boxes, respectively.

The DFL stabilizes and refines the regression of the bounding box coordinates $(x, y, w, h)$, serving as an auxiliary signal to the ProbIoU loss

$$L_{\text{dfl}(S_i, S_{i+1})} = -\left(\left(y_{i+1} - y\right) \log\left(S_i\right) + \left(y - y_i\right) \log\left(S_{i+1}\right)\right) \tag{17}$$

where $y_i$ and $y_{i+1}$ are the consecutive integer labels satisfying $y_i < y < y_{i+1}$, and $S_i$ and $S_{i+1}$ are the predicted probabilities from a Softmax layer, $S_i = (y_{i+1} - y)/(y_{i+1} - y_i)$ and $S_{i+1} = (y - y_i)/(y_{i+1} - y_i)$.

## IV. Experimental Results

### A. Datasets

The DroneVehicle dataset [9] is a large-scale drone-based vehicle detection dataset, comprising 28 439 paired RGB-IR images across five categories: car, truck, bus, van, and freight car. In accordance with the original publication's protocol, we partition the dataset into a training set with 17 990 pairs, a validation set with 1469 pairs, and a test set with 8980 pairs. We remove the white border with a width of 100 pixels in the original dataset image through preprocessing.

The VEDAI dataset [10] is another RGB-IR aerial data set for small vehicle detection, consisting of 1246 paired images. The data are split into training and testing sets following the officially provided methodology. While VEDAI is available in both $1024 \times 1024$ and $512 \times 512$ resolutions, we utilize the $512 \times 512$ images for our experiments. The dataset encompasses nine categories: car, truck, tractor, camping car, van, pickup, boat, plane, and others.

The OGSOD-1.0 [11] is a recently released SAR-optical paired image dataset for multi-object detection in remote sensing. It contains 14 665 training pairs and 3666 test pairs, with an original resolution of $256 \times 256$ pixels. The dataset includes over 48 000 annotated instances across three static categories: bridges, harbors, and oil tanks.

### B. Implementation Details

All experiments are conducted on an NVIDIA 4080 graphics processing unit (GPU), and the proposed model is implemented in PyTorch 2.1.0. For the DroneVehicle and VEDAI datasets, all input images are resized to a resolution of $640 \times 640$ pixels, and the model is trained for 300 epochs with batch sizes of 8 and 4. For the OGSOD-1.0, the images are resized to $512 \times 512$ with a batch size of 8 and trained for 100 epochs. The model is optimized using stochastic gradient descent (SGD) with an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 0.0005, following the default training configuration of YOLOv11 [18]. The training retains the default data augmentation protocol, including Mosaic with 1.0 probability and random horizontal flip with 0.5 probability. This identical protocol is consistently applied to all models within our ablation study to ensure a fair internal comparative analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS ON THE DRONEVEHICLE DATASET

| Detectors | Modality | Car | Freight car | Truck | Bus | Van | mAP@0.5 |
|---|---|---|---|---|---|---|---|
| RetinaNet [15] | RGB | 67.50 | 13.72 | 28.24 | 62.05 | 19.26 | 38.16 |
| Faster R-CNN [6] | | 67.88 | 26.31 | 38.59 | 66.98 | 23.20 | 44.59 |
| RoITransformer [27] | | 68.13 | 29.08 | 44.17 | 70.55 | 27.64 | 47.91 |
| Oriented R-CNN [28] | | 80.20 | 41.57 | 53.83 | 85.43 | 43.32 | 60.85 |
| S²A-Net [29] | | 80.03 | 42.24 | 54.17 | 84.92 | 43.81 | 61.03 |
| ReDet [30] | | 69.48 | 31.46 | 47.87 | 77.37 | 29.03 | 51.04 |
| YOLOv11-OBB [18] | | 93.70 | 51.90 | 64.50 | 91.30 | 51.70 | 70.60 |
| RetinaNet [15] | IR | 79.86 | 28.05 | 32.84 | 67.32 | 16.44 | 44.90 |
| Faster R-CNN [6] | | 88.63 | 35.16 | 42.51 | 77.92 | 28.52 | 54.55 |
| RoITransformer [27] | | 88.85 | 41.49 | 51.53 | 79.48 | 34.39 | 59.15 |
| Oriented R-CNN [28] | | 89.82 | 53.12 | 57.37 | 89.25 | 45.32 | 66.98 |
| S²A-Net [29] | | 89.93 | 55.81 | 54.52 | 88.87 | 48.42 | 67.51 |
| ReDet [30] | | 89.47 | 42.82 | 53.95 | 79.89 | 36.56 | 60.54 |
| YOLOv11-OBB [18] | | 98.30 | 66.90 | 77.70 | 95.20 | 61.20 | 79.90 |
| UA-CMDet [9] | RGB+IR | 87.51 | 46.80 | 60.70 | 87.08 | 37.95 | 64.01 |
| CIAN [57] | | 90.10 | 60.70 | 63.80 | 89.10 | 50.30 | 70.80 |
| AR-CNN [58] | | 90.10 | 62.10 | 64.80 | 89.40 | 51.50 | 71.60 |
| TSFADet [33] | | 89.88 | 63.74 | 67.87 | 89.81 | 53.99 | 73.06 |
| C²Former [34] | | 90.20 | 64.40 | 68.30 | 89.80 | 58.50 | 74.20 |
| CALNet [36] | | 90.30 | 62.97 | 76.15 | 89.11 | 58.46 | 75.39 |
| SLBAF-Net [59] | | 90.20 | 68.60 | 72.00 | 89.90 | 59.90 | 76.10 |
| CAGTDet [60] | | 90.82 | 66.28 | 69.65 | 90.46 | 55.62 | 74.57 |
| FMCFNet [61] | | 90.33 | 65.65 | 77.71 | 89.35 | 60.71 | 76.75 |
| FMPFNet [62] | | 95.18 | 60.47 | 76.79 | 94.46 | 57.08 | 76.80 |
| DMM+S²A-Net [63] | | 90.40 | 68.20 | 79.80 | 89.90 | 68.60 | 79.40 |
| DEPFusion [64] | | 90.50 | 66.60 | 79.40 | 90.00 | 67.80 | 78.90 |
| LPANet [65] | | 90.40 | 65.00 | 78.00 | 89.50 | 65.40 | 77.70 |
| CCLDet [37] | | 97.70 | 68.80 | 75.40 | 95.70 | 59.50 | 79.40 |
| Baseline | | 98.50 | 64.30 | 79.70 | 95.30 | 60.50 | 79.70 |
| CMFADet | | 98.60 | 71.60 | 82.90 | 96.20 | 67.80 | 83.42 |

### C. Evaluation Metrics

The model's performance is assessed using several standard metrics: precision ($P$), recall ($R$), $F1$-score, and mean average precision (mAP). For a comprehensive evaluation, we calculated mAP at different IoU thresholds, specifically reporting mAP@0.5 (IoU threshold of 0.5) and mAP@0.5:0.95 (averaged over IoU thresholds from 0.5 to 0.95 in steps of 0.05). Although the model is trained on all nine categories of the VEDAI dataset, the "plane" class is excluded from the mAP calculation due to its limited number of instances, which could otherwise skew the results.

### D. Comparison of Performance

To rigorously assess the efficacy of our proposed CMFADet method, we conduct a comprehensive series of comparative experiments against state-of-the-art object detection algorithms. We evaluate both single-modality methods for RGB or IR images, including RetinaNet, Faster R-CNN, RoITransformer, $S^2$A-Net, YOLOv11-OBB, and multimodality methods such as UA-CMDet, $C^2$Former, CALNet, SLBAF-Net, YOLOrs, as well as recently developed methods, including CAGTDet, FMCFNet, DMM, DEPFusion, and LPANet.

*1) Quantitative Numerical Results:* The comprehensive evaluation on the DroneVehicle and VEDAI datasets, presented in Tables I and II, showcases the performance of various object detectors across single (RGB or IR) and multimodality (RGB+IR) inputs. A clear and consistent trend observed on both datasets is the advantage of multimodal fusion over single-modality approaches.

On the DroneVehicle dataset, single-modality evaluations reveal the superiority of the IR modality over RGB. For example, YOLOv11-OBB's performance improved from a 70.60% mAP@0.5 on RGB to 79.90% on IR, highlighting the value of thermal information. By synergistically integrating both RGB and IR data, the proposed CMFADet method achieves an mAP@0.5 of 83.42%. This result not only substantially surpasses the best single-modality score but also outperforms all other multimodal methods. Notably, CMFADet exceeds the latest comparable methods, including LPANet (77.70%), DEPFusion (78.90%), FMCFNet (76.75%), FMPFNet (76.80%), and CAGTDet (74.57%). Specifically, CMFADet improves upon the next-best competitors, DMM with $S^2$A-Net (79.40%) and CCLDet (79.40%), by 4.02 percentage points. A detailed class-wise analysis further highlighted the model's exceptional performance, including car (98.60%) and bus (96.20%), and strong performance in challenging categories like truck (82.90%) and van (67.80%).

On the VEDAI dataset, the experimental results corroborate the superiority of our approach. Single-modality methods underperform relative to multimodal approaches, with the best RGB and IR modality (YOLOv8-OBB) achieving 69.65% and 61.40% mAP@0.5, respectively. Multimodal algorithms provide consistent benefits, where the CMFADet achieves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS ON THE VEDAI DATASET

| Detectors | Modality | Car | Truck | Tractor | Camping Car | Van | Pickup | Boat | Others | mAP@0.5 |
|---|---|---|---|---|---|---|---|---|---|---|
| RetinaNet [15] | RGB | 70.87 | 54.74 | 50.47 | 59.36 | 47.45 | 62.98 | 17.66 | 19.78 | 47.91 |
| Faster R-CNN [6] | | 71.40 | 54.20 | 61.00 | 70.50 | 59.50 | 67.60 | 52.30 | 40.10 | 59.58 |
| RoITransformer [27] | | 77.30 | 56.10 | 64.70 | 73.60 | 60.20 | 71.50 | 56.70 | 42.80 | 62.86 |
| S²A-Net [29] | | 74.50 | 47.30 | 55.60 | 61.70 | 32.50 | 65.10 | 16.70 | 39.80 | 49.15 |
| YOLOrs [66] | | 85.25 | 50.65 | 76.77 | 70.31 | 38.92 | 72.93 | 18.65 | 42.67 | 57.00 |
| YOLOv8-OBB [17] | | 88.30 | 73.60 | 69.00 | 67.20 | 77.20 | 75.40 | 59.60 | 46.90 | 69.65 |
| YOLOv11-OBB [18] | | 86.50 | 72.20 | 75.50 | 64.60 | 68.40 | 72.30 | 39.00 | 48.50 | 65.87 |
| RetinaNet [15] | IR | 59.66 | 52.81 | 35.30 | 53.38 | 42.29 | 54.09 | 9.57 | 24.89 | 41.50 |
| Faster R-CNN [6] | | 71.60 | 49.10 | 49.20 | 68.10 | 57.00 | 66.50 | 35.60 | 29.50 | 53.33 |
| RoITransformer [27] | | 76.10 | 51.70 | 51.90 | 71.20 | 64.30 | 70.70 | 46.90 | 28.30 | 57.64 |
| S²A-Net [29] | | 73.00 | 39.20 | 41.90 | 59.20 | 32.30 | 65.60 | 13.90 | 23.10 | 43.52 |
| YOLOrs [66] | | 82.03 | 54.21 | 54.39 | 63.80 | 43.38 | 73.92 | 21.97 | 43.99 | 54.71 |
| YOLOv8-OBB [17] | | 83.50 | 65.80 | 43.10 | 58.70 | 85.80 | 73.60 | 46.00 | 34.70 | 61.40 |
| YOLOv11-OBB [18] | | 85.90 | 61.90 | 55.00 | 66.10 | 59.30 | 72.60 | 40.50 | 30.50 | 59.98 |
| C²Former+S²A-Net [34] | RGB+IR | 81.20 | 59.40 | 37.70 | 77.60 | 53.70 | 78.30 | 52.60 | 46.10 | 60.82 |
| YOLOrs [66] | | 84.15 | 52.60 | 67.88 | 68.81 | 57.91 | 78.27 | 21.47 | 46.75 | 59.73 |
| Oriented R-CNN+CMAFF [67] | | 81.70 | 58.80 | 68.70 | 78.40 | 68.50 | 76.30 | 66.00 | 51.50 | 68.74 |
| CALNet [36] | | 82.30 | 66.70 | 73.00 | 75.40 | 65.50 | 78.20 | 56.60 | 47.00 | 68.09 |
| DMM+S²A-Net [63] | | 77.90 | 59.30 | 68.10 | 70.80 | 57.40 | 75.80 | 61.20 | 43.50 | 64.25 |
| DMM+Oriented R-CNN [63] | | 84.20 | 65.70 | 72.30 | 79.00 | 72.50 | 78.80 | 72.30 | 56.20 | 72.63 |
| LPANet [65] | | 82.40 | 70.60 | 77.00 | 78.40 | 75.40 | 80.00 | 69.00 | 56.70 | 73.69 |
| Baseline | | 88.60 | 73.30 | 69.90 | 67.50 | 60.10 | 77.10 | 61.00 | 58.60 | 69.51 |
| CMFADet | | 88.20 | 85.30 | 79.60 | 77.60 | 77.10 | 78.20 | 62.80 | 61.40 | 76.28 |

76.28% mAP@0.5, surpassing the baseline (69.51%) and the oriented R-CNN with CMAFF (68.74%), by 6.77 and 7.54 percentage points. Additionally, CMFADet performs better than these methods, including DMM with $S^2$A-Net (64.25%) and with oriented R-CNN (72.63%) and LPANet (73.69%). CMFADet also delivers top AP scores in most categories, such as truck (85.30%), van (77.10%), and others (61.40%), indicating strong generalization across diverse object types.

To demonstrate the generality of CMFADet, we evaluate it on the optical. SAR OGSOD-1.0 dataset (Table III). Single-modality results show that the YOLOv8-OBB achieved 81.0% mAP@0.5 on optical domain, while its performance dropped to 73.7% mAP@0.5 on SAR domain. Despite the large domain shift from RGB-IR, CMFADet outperforms the prior multimodal fusion and knowledge distillation methods with the best mAP@0.5 of 85.2%. The performance is particularly evident in the challenging oil tank category, where CMFADet achieves an AP of 80.3%, a drastic improvement over MO R-CNN (71.4% AP). CMFADet also exhibits highly competitive performance on harbor (92.7%). These results validate the strong feature adaptation capability of the CMFADet architecture across diverse sensor modalities.

Overall, these results demonstrate that CMFADet consistently outperforms both conventional and recent state-of-the-art multimodality detectors, highlighting the effectiveness of its cross-modality feature adaptive design.

*2) Qualitative Visualization Results:* In addition to the quantitative analysis, a qualitative comparison of the detection performance on the DroneVehicle and VEDAI datasets is presented in Figs. 7 and 8. As shown in Fig. 7, CMFADet exhibits robust performance under challenging conditions, including low-light environments, dense vehicle distributions, and occlusions by plants. Competing methods often suffer from missed detections or false alarms due to insufficient multimodal feature exploitation. For instance, in low-light scenes, CMFADet reduces false alarms and stabilizes detections. In cases of dense vehicle distribution, other methods show a certain degree of missed detections or incorrect merging, while the CMFADet accurately distinguishes and localizes each individual vehicle. The ability to robustly identify

TABLE III
COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS ON THE OGSOD-1.0 DATASET

| Detectors | Modality | Bridge | Harbor | Oil Tank | mAP@0.5 |
|---|---|---|---|---|---|
| R3Det [68] | Optical | 76.8 | 90.8 | 59.6 | 75.7 |
| Oriented R-CNN [28] | | 88.7 | 90.8 | 62.0 | 80.5 |
| S²A-Net [29] | | 77.2 | 90.7 | 69.3 | 79.1 |
| KFIoU [69] | | 89.2 | 90.8 | 57.5 | 79.2 |
| YOLOv8-OBB [17] | | 80.7 | 91.7 | 70.5 | 81.0 |
| YOLOv11-OBB [18] | | 76.3 | 88.5 | 77.5 | 80.8 |
| R3Det [68] | SAR | 58.1 | 90.7 | 31.8 | 60.2 |
| Oriented R-CNN [28] | | 68.6 | 90.2 | 40.6 | 66.4 |
| S²A-Net [29] | | 66.8 | 86.9 | 52.4 | 68.7 |
| KFIoU [69] | | 68.7 | 89.8 | 30.8 | 63.1 |
| YOLOv8-OBB [17] | | 67.9 | 90.7 | 62.3 | 73.7 |
| YOLOv11-OBB [18] | | 63.8 | 87.3 | 59.2 | 70.1 |
| C²Former [34] | Optical + SAR | 77.1 | 87.5 | 64.8 | 76.5 |
| CALNet [36] | | 73.7 | 64.6 | 63.2 | 67.2 |
| CoLD [11] | | 74.8 | 98.0 | 31.4 | 68.0 |
| DDCINet+RoITans [70] | | - | - | - | 80.8 |
| DDCINet+Swin-T [70] | | - | - | - | 81.1 |
| MO R-CNN [71] | | 88.8 | 90.5 | 71.4 | 83.6 |
| Baseline | | 81.2 | 91.0 | 79.0 | 83.7 |
| CMFADet | | 82.5 | 92.7 | 80.3 | 85.2 |

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

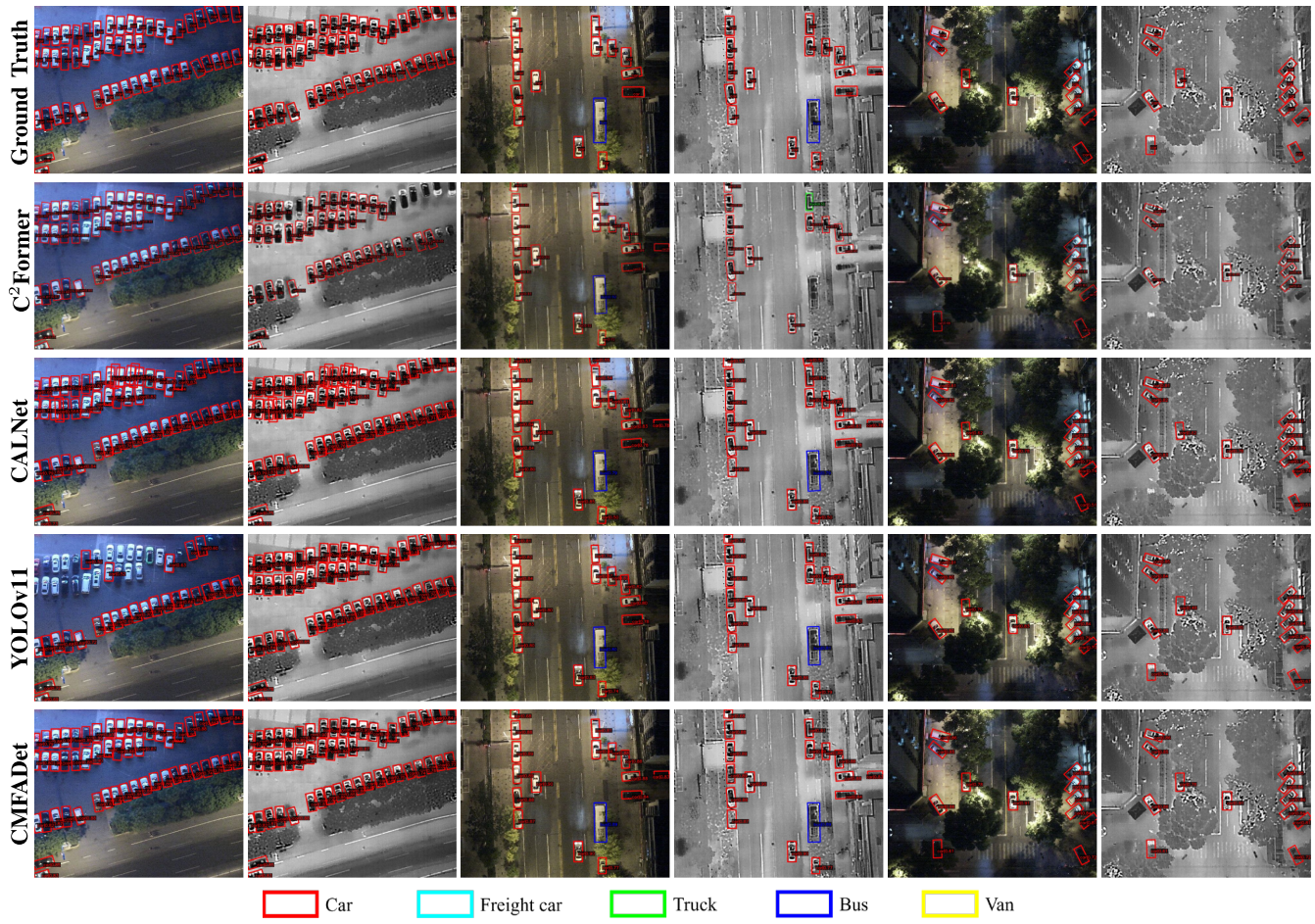



Fig. 7. Qualitative comparison of our proposed CMFADet with C$^2$Former, CALNet, and YOLOv11 on the DroneVehicle dataset. The displayed bounding boxes have an IoU greater than 0.5, and a corresponding color legend is provided at the bottom of the figure.

partially visible targets in scenes with plant occlusion further demonstrates the superior capability of our method in handling such complex cases. On the VEDAI dataset, which features tiny vehicles with ambiguous visual features, other methods struggle with localization and recognition, frequently missing targets. The CMFADet method consistently delivers accurate bounding boxes and stable detections, confirming its advantage in fusing multimodality features and explaining its superior mAP-scores.

*3) Computational Efficiency Analysis:* The computational efficiency of CMFADet is evaluated against recent multimodality object detectors on the DroneVehicle dataset in Table IV, comparing model parameters (Params), floating-point operations (FLOPs), inference speed (FPS), and detection accuracy (mAP@0.5). CMFADet achieves the superior tradeoff between accuracy and efficiency. CMFADet demonstrates ultralow complexity (3.64 M Params and 5.63 GFLOPs), making it comparable to the simplest Baseline model in terms of scale. Crucially, CMFADet attains an mAP@0.5 of 83.42%, effectively surpassing heavy-weight multimodality models such as FMCFNet (200.43 GFLOPs) and CCLDet (82.28 M Params and 293.35 GFLOPs). The model's strong inference speed of 108.7 FPS confirms its suitability for real-time UAV applications. While CMFADet introduces a marginal increase in computational cost over the Baseline detector (+0.82 GFLOPs), this small investment yields a substantial gain of +3.72% in mAP@0.5.

TABLE IV
PERFORMANCE AND EFFICIENCY COMPARISON OF RECENT METHODS ON THE DRONEVEHICLE DATASET

| Detectors | Input size | Params (M) | FLOPs (G) | FPS | mAP@0.5 |
|---|---|---|---|---|---|
| UA-CMDet [9] | 640 × 512 | 138.69 | 541.57 | 2.7 | 64.01 |
| CIAN [57] | 640 × 640 | - | 70.36 | 21.7 | 70.80 |
| AR-CNN [58] | 640 × 640 | - | 104.30 | 18.2 | 71.60 |
| TSFADet [33] | 640 × 512 | 104.70 | 109.80 | 18.6 | 73.06 |
| C²Former [34] | 640 × 512 | 100.80 | 89.90 | 30.0 | 74.20 |
| CALNet [36] | 640 × 640 | - | 174.54 | 18.9 | 75.39 |
| SLBAF-Net [59] | 640 × 640 | - | - | 63.2 | 76.10 |
| CAGTDet [60] | 640 × 640 | - | 120.60 | 17.8 | 74.57 |
| FMCFNet [61] | 640 × 640 | - | 200.43 | - | 76.75 |
| FMPFNet [62] | 640 × 640 | - | 26.30 | 132.0 | 76.80 |
| DMM+S²A-Net [63] | 640 × 512 | 87.97 | 108.80 | - | 79.40 |
| DEPFusion [64] | 640 × 512 | 80.30 | 108.30 | - | 78.90 |
| LPANet [65] | 640 × 640 | - | 184.63 | 36.1 | 77.70 |
| CCLDet [37] | 640 × 512 | 82.28 | 293.35 | 6.7 | 79.40 |
| Baseline | 640 × 640 | 3.62 | 4.81 | 172.4 | 79.70 |
| CMFADet | 640 × 640 | 3.64 | 5.63 | 108.7 | 83.42 |

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

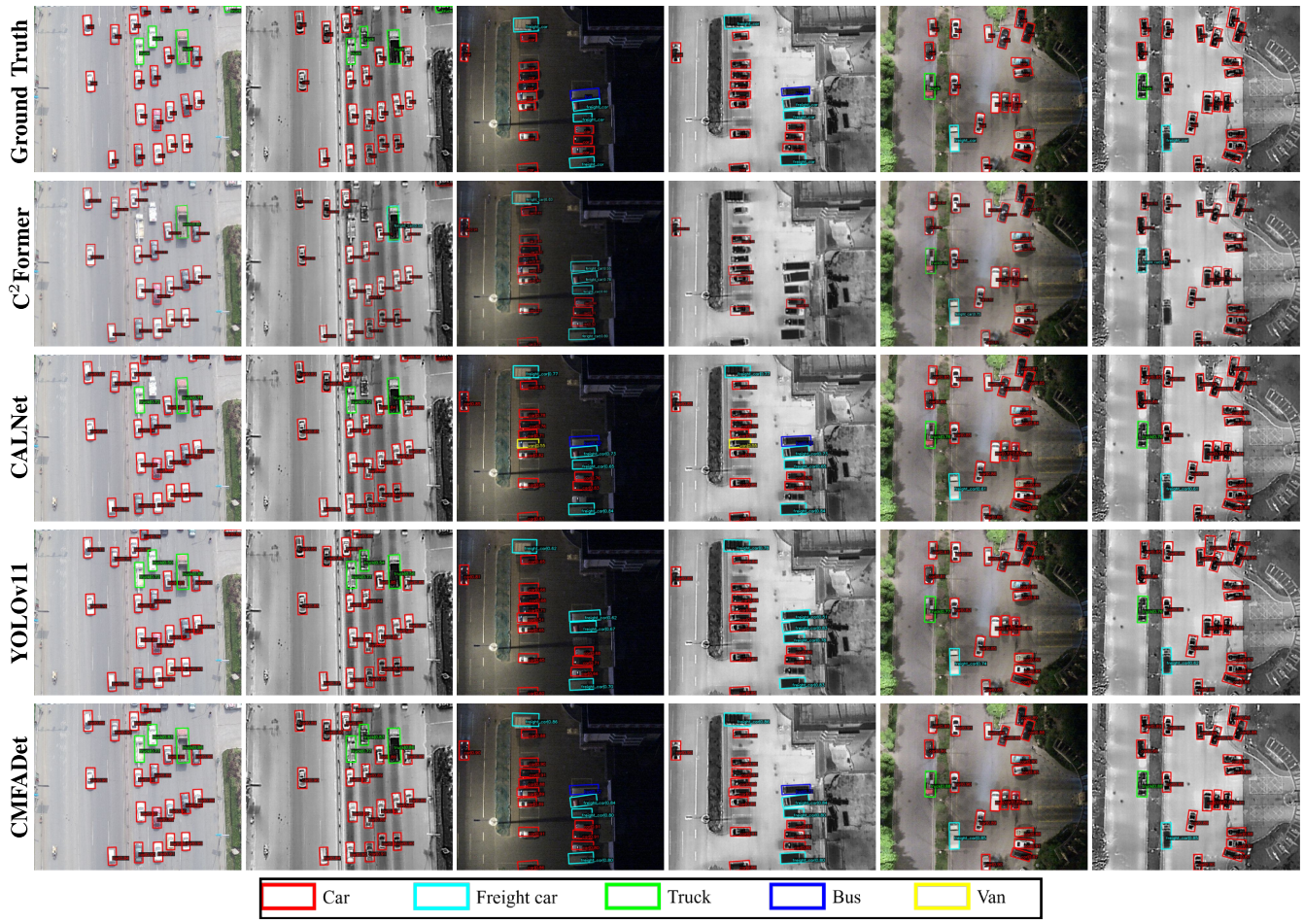



Fig. 7. *(Continued.)*Qualitative comparison of our proposed CMFADet with C$^2$Former, CALNet, and YOLOv11 on the DroneVehicle dataset. The displayed bounding boxes have an IoU greater than 0.5, and a corresponding color legend is provided at the bottom of the figure.

TABLE V
ABLATION EXPERIMENTS OF THE PROPOSED MODULES ON THE DRONEVEHICLE AND VEDAI DATASETS

| Method | SFEM | IR-AFAB | CIF | ATAH | GFLOPs | Params | DroneVehicle | | | | | VEDAI | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | $P$ | $R$ | $F1$ | $mAP_{50}$ | $mAP_{50:95}$ | $P$ | $R$ | $F1$ | $mAP_{50}$ | $mAP_{50:95}$ |
| Baseline | - | - | - | - | 4.82 | 3.62 | 77.3 | 76.3 | 76.8 | 79.7 | 64.4 | 75.8 | 59.0 | 66.4 | 69.5 | 49.2 |
| (a) | ✓ | - | - | - | 4.90 | 4.01 | 77.8 | 77.5 | 77.6 | 80.9 | 65.1 | 71.8 | 66.2 | 68.9 | 71.6 | 50.1 |
| (b) | - | ✓ | - | - | 4.89 | 4.02 | 77.9 | 78.2 | 78.0 | 81.4 | 65.7 | 67.2 | 62.9 | 65.0 | 69.3 | 49.6 |
| (c) | - | - | ✓ | - | 4.90 | 4.06 | 79.5 | 79.2 | 79.3 | 82.5 | 67.2 | 64.3 | 69.0 | 66.6 | 71.7 | 51.1 |
| (d) | - | - | - | ✓ | 5.65 | 3.60 | 79.3 | 79.0 | 79.1 | 81.9 | 67.5 | 72.3 | 61.1 | 66.2 | 72.3 | 51.3 |
| (e) | ✓ | ✓ | - | - | 4.88 | 4.01 | 77.6 | 78.3 | 77.9 | 80.9 | 65.3 | 72.2 | 62.8 | 67.1 | 71.9 | 51.8 |
| (f) | ✓ | ✓ | ✓ | - | 4.88 | 4.05 | 79.9 | 79.8 | 79.8 | 82.9 | 67.9 | 75.6 | 60.8 | 67.4 | 74.3 | 53.6 |
| (g) | ✓ | ✓ | ✓ | ✓ | 5.63 | 3.64 | 80.8 | 80.3 | 80.5 | 83.4 | 69.1 | 78.7 | 62.5 | 69.7 | 76.3 | 54.9 |

### *E. Ablation Study*

To systematically evaluate the contribution and generalizability of each proposed module, we conduct a series of ablation experiments on DroneVehicle and VEDAI datasets. The baseline model is built upon the YOLOv11-Nano (YOLOv11n [18]) architecture, serving as a robust and modern comparison reference. It adopts the default YOLOv11 training configuration and data augmentation settings to ensure a fair and rigorous comparison across all variants as detailed in Subsection IV-B. This setup allows us to isolate the performance gains brought by each module with high reliability. Table V summarizes the positive impact of each module, while Table VI provides a detailed breakdown of the internal configurations of the CIF module. The baseline model achieves an $mAP_{50}$ of 79.7% and an $mAP_{50:95}$ of 64.4% on DroneVehicle and an $mAP_{50}$ of 69.5% and an $mAP_{50:95}$ of 49.2% on VEDAI, establishing a high benchmark for subsequent modules.

The DroneVehicle dataset features complex, dense traffic scenes. The ATAH module proves most effective in isolation, achieving an $mAP_{50:95}$ of 67.5%, highlighting its capability in geometric alignment and refining localization quality for dense objects. The CIF module records a high $mAP_{50}$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

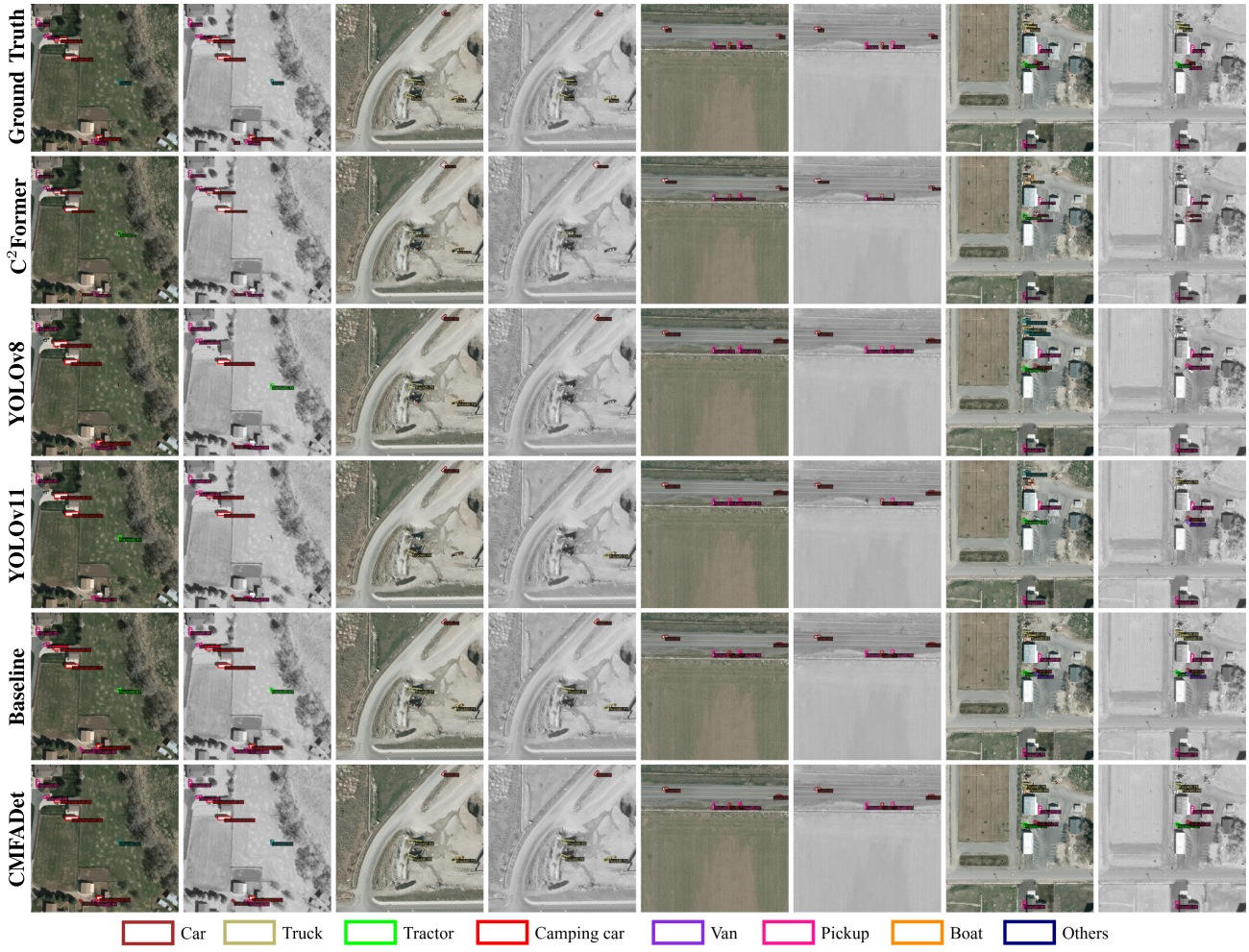



Fig. 8. Qualitative comparison of the proposed CMFADet with C$^2$Former, YOLOv8, and YOLOv11 on the VEDAI dataset. The displayed bounding boxes have an IoU greater than 0.5, and a corresponding color legend is provided at the bottom of the figure.

TABLE VI

ABLATION EXPERIMENTS OF THE CIF MODULE ON THE DRONEVEHICLE AND VEDAI DATASETS

| Data | Configuration | | | | Performance Metrics (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Method | Cross-sum | Attention | Stage | $P$ | $R$ | $F1$ | $mAP_{50}$ | $mAP_{50:95}$ |
| DroneVehicle | Baseline | - | - | - | 77.3 | 76.3 | 76.8 | 79.7 | 64.4 |
| | (c-1) | - | SE | $P_{3,4,5}$ | 78.7 | 78.4 | 78.5 | 81.7 | 66.5 |
| | (c-2) | ✓ | SE | $P_{3,4,5}$ | 79.5 | 79.2 | 79.3 | 82.5 | 67.2 |
| | (c-3) | ✓ | SE | $P_{4,5}$ | 79.6 | 78.4 | 79.0 | 82.1 | 66.9 |
| | (c-4) | ✓ | SE | $P_5$ | 79.0 | 78.7 | 78.8 | 81.9 | 66.5 |
| | (c-5) | ✓ | Spatial | $P_{3,4,5}$ | 78.2 | 78.6 | 78.4 | 81.8 | 66.4 |
| | (c-6) | ✓ | CBAM | $P_{3,4,5}$ | 78.7 | 78.4 | 78.5 | 81.7 | 66.6 |
| VEDAI | Baseline | - | - | - | 75.8 | 59.0 | 66.4 | 69.5 | 49.2 |
| | (c-1) | - | SE | $P_{3,4,5}$ | 65.8 | 61.1 | 63.4 | 68.5 | 48.6 |
| | (c-2) | ✓ | SE | $P_{3,4,5}$ | 64.3 | 69.0 | 66.6 | 71.7 | 51.1 |
| | (c-3) | ✓ | SE | $P_{4,5}$ | 67.5 | 68.1 | 67.8 | 68.5 | 48.2 |
| | (c-4) | ✓ | SE | $P_5$ | 69.6 | 57.7 | 63.1 | 68.6 | 49.4 |
| | (c-5) | ✓ | Spatial | $P_{3,4,5}$ | 69.4 | 65.0 | 67.1 | 71.0 | 50.9 |
| | (c-6) | ✓ | CBAM | $P_{3,4,5}$ | 63.3 | 66.2 | 64.7 | 71.0 | 51.4 |

of 82.5% among the enhancement group, indicating multi-scale feature aggregation. The SFEM and IR-AFAB modules both contribute to initial feature gains, pushing the mAP$_{50}$ past 80%. The proposed CMFADet, integrating these four modules, demonstrates powerful synergy. The final mAP$_{50}$ reaches 83.4% and mAP$_{50:95}$ reaches an exceptional 69.1%, representing a significant 4.7 percentage gain over the competitive baseline (64.4%).

The VEDAI dataset provides a different challenging profile, characterized by smaller object sizes and diverse aerial views. The SFEM, focusing on RGB feature stabilization, shows the substantial initial improvement in recall $R$ from 59.0% to 66.2%. The IR-AFAB module exhibits a minor tradeoff where $R$ improves from 59.0% to 62.9% at the temporary cost of $P$ and $F1$-score. The mAP$_{50:95}$ increases from 49.2% to 49.6%, validating its role in refining IR feature representation and establishing a necessary foundation for the final architecture. The CIF module, achieving the highest $R$ (69.0%), confirms the critical role of multiscale feature fusion in enhancing perception across different object sizes, reaching mAP$_{50}$ of 71.7% and mAP$_{50:95}$ of 51.1%. The ATAH module focuses on refining detection precision and quality, achieving the highest $P$ of 72.3% among all single modules, and also records the best mAP$_{50}$ and mAP$_{50:95}$ values at 72.3% and 51.3%. When submodules are integrated, the CMFADet achieves a $P$ of 78.7% and an $F1$-score of 69.7%, with metrics soaring to mAP$_{50}$ of 76.3% and mAP$_{50:95}$ of 54.9%. This indicates a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

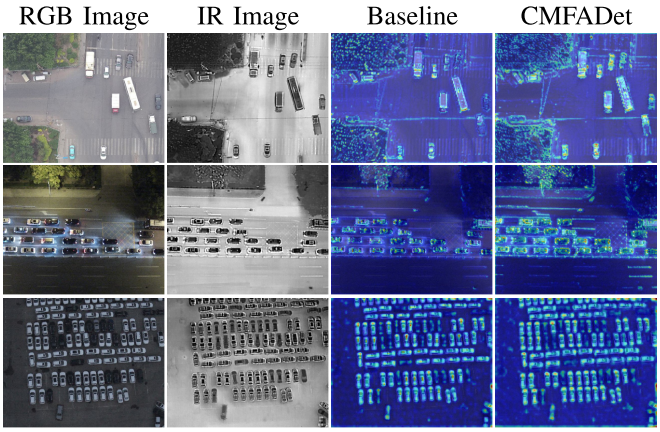



Fig. 9. Visualization of feature maps on the DroneVehicle dataset.

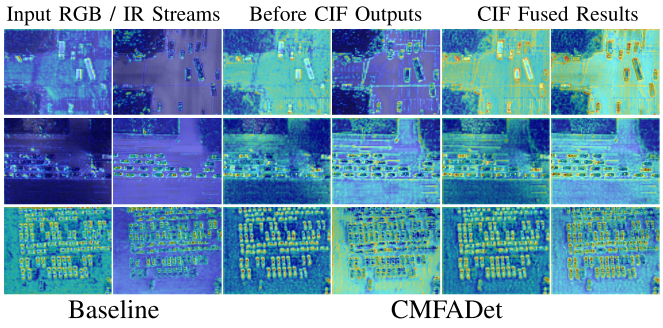



Fig. 10. Visualization of CIF module on the DroneVehicle dataset.

powerful synergistic effect where the combination of SFEM, IR-AFAB, CIF, and ATAH modules leads to a substantial improvement across performance metrics.

Table VI provides a detailed analysis of the internal configurations of the CIF module. The experiments comparing (c-2)–(c-4) reveal that applying the CIF module across all three feature stages ($P_3$–$P_5$) is optimal. Limiting fusion to later stages, such as $P_4$ and $P_5$ (c-3), $\text{mAP}_{50}$ drops to 68.5% on VEDAI, falling below the baseline. Further restricting fusion to only $P_5$ (c-4) results in even greater performance degradation. This clearly demonstrates that fusing fine-grained features from the early $P_3$ stage is crucial for accurate object detection. Furthermore, we assess various attention mechanisms. While the convolutional block attention module (CBAM) [73] has a slightly greater $\text{mAP}_{50:95}$ of 51.4% on VEDAI, the SE attention [72] (c-2) is selected as the mechanism. This choice is justified by its consistently higher $\text{mAP}_{50}$ on both DroneVehicle (82.5%) and VEDAI (71.7%), providing its optimal overall balance and a higher $F1$-score on DroneVehicle (79.3%). By comparing (c-2) with (c-1), the cross-sum operation is central to the CIF module's performance for integrating features from different levels and ensuring an effective information flow.

Fig. 9 intuitively visualizes the sample feature maps between the CMFADet and the baseline model on DroneVehicle. The heatmaps generated by the baseline model are often dispersed and blurred, failing to precisely focus on individual targets, especially when they are dense or small. In contrast, the feature maps from the CMFADet model exhibit much more concentrated and accurate activation regions. In the aerial traffic scene shown in the first row, CMFADet precisely identifies each vehicle, including both large buses and smaller cars. In the dense parking lot scenes of the second and third rows, while the baseline model produces large, indistinct areas of activation, CMFADet successfully highlights each vehicle individually. This demonstrates our method's superior discrimination and localization capabilities when dealing with dense, small objects.

In addition, Fig. 10 visualizes the effect of the CIF module through the fused feature heatmaps. The first two columns present the input RGB and IR features from the baseline model, which are the most dispersed and least focused. The subsequent columns compare the feature representation before and after the CIF module's operations. The features in the third and fourth columns, which represent the results without cross-modality enhancement, are typically fragmented, widely distributed, and weak. The CIF fused results incorporate the CIF module's cross-modality residual interaction, exhibiting activation that is more concentrated, stronger in intensity, and sharply bounded around the targets. This visualization confirms the CIF module's role in actively mitigating the detrimental effect of feature misalignment and enhancing target-specific information during the fusion process.

### F. Discussion

The superior performance of CMFADet stems from a systematic framework that effectively exploits multimodal information for vehicle detection. The method enhances modality-specific features prior to fusion and employs a dynamic fusion strategy. As quantitative and qualitative results confirm, CMFADet achieves a substantial performance advantage over state-of-the-art and baseline methods. Specifically, CMFADet records the highest mAP@0.5 of 83.42% on the DroneVehicle dataset, surpassing the second-best method by 4.02 percentage points. This superiority is confirmed across different challenge profiles, including VEDAI and the OGSOD-1.0 datasets, validating the architecture's strong cross-modality generalization capability. The effectiveness of each component (SFEM, IR-AFAB, CIF, and ATAH) is confirmed through the ablation study presented in Table V. From a modeling perspective, SFEM is designed as a unified spatial–frequency feature enhancement module that complements conventional spatial feature learning. The ablation results demonstrate that incorporating frequency-domain feature transformations alongside spatial edge-aware information consistently improves detection performance across datasets, validating the effectiveness of the spatial–frequency design in complex aerial scenarios.

Despite its strong performance in general scenarios, CMFADet still has room for improvement. For instance, while the model excels at detecting well-defined classes such as car and bus on the DroneVehicle dataset, its performance is relatively lower for fine-grained categories like truck and freight car. This confusion is mainly caused by the top-down aerial viewpoint, where discriminative structural differences are weakened by scale variation, limited spatial resolution, and partial occlusion. Our computational analysis (Table IV) confirms the model achieves a superior accuracy–efficiency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

tradeoff compared to SOTA methods, validating its suitability for aerial deployment. While CMFADet is already highly efficient, future research will explore lightweight model variants or pruning techniques to further reduce computational cost. Further extensions will include vehicle tracking and video-based detection to broaden the framework's applicability in real-world scenarios.

## V. Conclusion

In this article, we present CMFADet, a novel cross-modality detection framework tailored for aerial RGB-IR imagery under challenging conditions. To address modality imbalance and misalignment, we propose a multimodal backbone incorporating an SFEM and an IR-AFAB for dedicated feature extraction, a CIF module for adaptive cross-modality interaction, and an ATAH head for task-aware feature alignment. Extensive experiments on DroneVehicle, VEDAI, and OGSOD-1.0 demonstrate that CMFADet achieves consistent performance gains over state-of-the-art methods with favorable computational efficiency, particularly in low-light and small-object scenarios. Overall, this work highlights the importance of maintaining modality-specific features and task-specific representations in multimodal detection. Future research may explore extending CMFADet to real-time or sequential aerial perception tasks.

## References


[1] Z.-Q. Zhao, P. Zheng, S.-T. Xu, and X. Wu, "Object detection with deep learning: A review," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 11, pp. 3212–3232, Nov. 2019.

[2] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proc. IEEE*, vol. 111, no. 3, pp. 257–276, Mar. 2023.

[3] S. Gui, S. Song, R. Qin, and Y. Tang, "Remote sensing object detection in the deep learning era—A review," *Remote Sens.*, vol. 16, no. 2, p. 327, Jan. 2024.

[4] P. Zhu et al., "Detection and tracking meet drones challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, Nov. 2022.

[5] D. Du et al., "The unmanned aerial vehicle benchmark: Object detection and tracking," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 370–386.

[6] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[8] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[9] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022.

[10] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.

[11] C. Wang, R. Ruan, Z. Zhao, C. Li, and J. Tang, "Category-oriented localization distillation for SAR object detection and a unified benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5211314, doi: 10.1109/TGRS.2023.3291356.

[12] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[13] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[14] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*. Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 21–37.

[15] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.

[16] G. Jocher et al., "ultralytics/yolov5: v3.0," 2020. [Online]. Available: https://github.com/ultralytics/yolov5/tree/v3.0

[17] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *Ultralytics YOLO (Version 8.0.0)*. [Online]. Available: https://github.com/ultralytics/ultralytics

[18] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.

[19] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[20] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[21] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[22] X. Dai et al., "Dynamic head: Unifying object detection heads with attentions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Apr. 2021, pp. 7373–7382.

[23] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets V2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.

[24] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3490–3499.

[25] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2024, pp. 16965–16974.

[26] C. Yu and Y. Shin, "MCG-RTDETR: Multi-convolution and context-guided network with cascaded group attention for object detection in unmanned aerial vehicle imagery," *Remote Sens.*, vol. 16, no. 17, p. 3169, Aug. 2024.

[27] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[28] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[29] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[30] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[31] C. Song, Y. Zhang, M. Ma, and S. Mei, "RIFormer+: Rethinking rotation-invariant feature learning in transformer," *IEEE Trans. Multimedia*, vol. 27, pp. 8423–8435, 2025.

[32] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[33] M. Yuan, Y. Wang, and X. Wei, "Translation, scale and rotation: Cross-modal alignment meets RGB-infrared vehicle detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 509–525.

[34] M. Yuan and X. Wei, "$C^2$former: Calibrated and complementary transformer for RGB-infrared object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403712, doi: 10.1109/TGRS.2024.3376819.

[35] C. Bao, J. Cao, Q. Hao, Y. Cheng, Y. Ning, and T. Zhao, "Dual-YOLO architecture from infrared and visible images for object detection," *Sensors*, vol. 23, no. 6, p. 2934, Mar. 2023.

[36] X. He, C. Tang, X. Zou, and W. Zhang, "Multispectral object detection via cross-modal conflict-aware learning," in *Proc. 31st ACM Int. Conf. Multimedia*, Ottawa, ON, Canada, Oct. 2023, pp. 1465–1474.

[37] X. Shang et al., "CCLDet: A cross-modality and cross-domain low-light detector," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 3, pp. 3284–3294, Mar. 2025.

[38] J. Nie, H. Sun, X. Sun, L. Ni, and L. Gao, "Cross-modal feature fusion and interaction strategy for CNN-transformer-based object detection in visual and infrared remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[39] H. Tang, Z. Li, D. Zhang, S. He, and J. Tang, "Divide-and-conquer: Confluent triple-flow network for RGB-T salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1958–1974, Mar. 2025.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.

[40] F. Sun, P. Ren, B. Yin, F. Wang, and H. Li, "CATNet: A cascaded and aggregated transformer network for RGB-D salient object detection," *IEEE Trans. Multimedia*, vol. 26, pp. 2249–2262, 2024.
[41] X. Hu, F. Sun, J. Sun, F. Wang, and H. Li, "Cross-modal fusion and progressive decoding network for RGB-D salient object detection," *Int. J. Comput. Vis.*, vol. 132, no. 8, pp. 3067–3085, Aug. 2024.
[42] Y. Quan, D. Zhang, and J. Tang, "Generalized concordant vision transformer with masked image tokens for object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 11, pp. 10616–10631, Nov. 2025.
[43] Y. Duan, X. Wu, H. Deng, and L.-J. Deng, "Content-adaptive non-local convolution for remote sensing pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27738–27747.
[44] Y. Yao, Y. Zhang, Y. Wan, X. Liu, X. Yan, and J. Li, "Multi-modal remote sensing image matching considering co-occurrence filter," *IEEE Trans. Image Process.*, vol. 31, pp. 2584–2597, 2022.
[45] Y. Zhang, S. Yan, L. Zhang, and B. Du, "Fast projected fuzzy clustering with anchor guidance for multimodal remote sensing imagery," *IEEE Trans. Image Process.*, vol. 33, pp. 4640–4653, 2024.
[46] J. Zheng, Z. Li, P. Duan, X. Kang, and W. Fu, "Hyperspectral remote sensing scene classification with spectral–spatial convolutional network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Athens, Greece, Jul. 2024, pp. 9464–9467.
[47] X. Zhang, Y. Feng, N. Wang, G. Lu, and S. Mei, "Aerial person detection for search and rescue: Survey and benchmarks," *J. Remote Sens.*, vol. 5, p. 474, Jan. 2025.
[48] X. Zhang, Y. Feng, S. Zhang, N. Wang, G. Lu, and S. Mei, "Robust aerial person detection with lightweight distillation network for edge deployment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5630616.
[49] X. Zhang, Y. Feng, N. Wang, G. Lu, and S. Mei, "Transformer-based person detection in paired RGB-T aerial images with VTSaR dataset," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5082–5099, 2025.
[50] B. Jähne, H. Haussecker, and P. Geissler, *Handbook of Computer Vision and Applications*. New York, NY, USA: Academic, 1999.
[51] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10819–10829.
[52] H. Touvron, M. Cord, A. Sablayrolles, G. Synnaeve, and H. Jégou, "Going deeper with image transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 32–42.
[53] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger, "Deep networks with stochastic depth," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 646–661.
[54] D. Shi, "TransNeXt: Robust foveal visual perception for vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2024, pp. 17773–17783.
[55] X. Li et al., "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. Annu. Conf. Neural Inf. Process. Syst. (NIPS)*, Dec. 2020, pp. 21002–21012.
[56] J. M. Llerena, L. Felipe Zeni, L. N. Kristen, and C. Jung, "Gaussian bounding boxes and probabilistic intersection-over-union for object detection," 2021, *arXiv:2106.06072*.
[57] L. Zhang et al., "Cross-modality interactive attention network for multispectral pedestrian detection," *Inf. Fusion*, vol. 50, pp. 20–29, Oct. 2019.
[58] L. Zhang, X. Zhu, X. Chen, X. Yang, Z. Lei, and Z. Liu, "Weakly aligned cross-modal learning for multispectral pedestrian detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5127–5137.
[59] X. Cheng, K. Geng, Z. Wang, J. Wang, Y. Sun, and P. Ding, "SLBAF-Net: Super-lightweight bimodal adaptive fusion network for UAV detection in low recognition environment," *Multimedia Tools Appl.*, vol. 82, no. 30, pp. 47773–47792, Dec. 2023.
[60] M. Yuan, X. Shi, N. Wang, Y. Wang, and X. Wei, "Improving RGB-infrared object detection with cascade alignment-guided transformer," *Inf. Fusion*, vol. 105, May 2024, Art. no. 102246.
[61] Y. Liu and W. Jiang, "Frequency mining and complementary fusion network for RGB-infrared object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[62] Z. Wang and Q. Zhang, "Real-time aerial multispectral object detection with dynamic modality-balanced pixel-level fusion," *Sensors*, vol. 25, no. 10, p. 3039, May 2025.
[63] M. Zhou et al., "DMM: Disparity-guided multispectral Mamba for oriented object detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5404913.
[64] S. Li, Z. Liu, Z. Hong, Z. Zhou, and X. Cao, "DEPFusion: Dual-domain enhancement and priority-guided mamba fusion for UAV multispectral object detection," 2025, *arXiv:2509.07327*.
[65] W. Wu, C. Li, X. Wang, B. Luo, and Q. Liu, "Large language model guided progressive feature alignment for multimodal UAV object detection," 2025, *arXiv:2503.06948*.
[66] M. Sharma et al., "YOLOrs: Object detection in multimodal remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1497–1508, 2021.
[67] Q. Fang and Z. Wang, "Cross-modality attentive feature fusion for object detection in multispectral remote sensing imagery," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108786.
[68] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2021, vol. 35, no. 4, pp. 3163–3171.
[69] X. Yang et al., "The KFIoU loss for rotated object detection," 2022, *arXiv:2201.12558*.
[70] W. Bao, M. Huang, J. Hu, and X. Xiang, "Dual-dynamic cross-modal interaction network for multimodal remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5401013, doi: 10.1109/TGRS.2025.3530085.
[71] L. Wang et al., "MO R-CNN: Multispectral oriented R-CNN for object detection in remote sensing image," 2025, *arXiv:2509.16957*.
[72] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7132–7141.
[73] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, 2018, pp. 3–19.



**Chushi Yu** (Graduate Student Member, IEEE) received the B.S. degree from Hangzhou Normal University, Hangzhou, China, in 2017, and the M.S. degree from Soongsil University, Seoul, South Korea, in 2022, where she is currently pursuing the Ph.D. degree with the School of Electronic Engineering.

Her research interests include synthetic aperture radar, computer vision, and multimodal learning.

Ms. Yu was a recipient of the 2025 ICT Express Best Paper Award from KICS.



**Yoan Shin** (Senior Member, IEEE) received the B.S. and M.S. degrees in electronics engineering from Seoul National University, Seoul, South Korea, in 1987 and 1989, respectively, and the Ph.D. degree in electrical and computer engineering from The University of Texas at Austin, Austin, TX, USA, in 1992.

From 1992 to 1994, he was a Member of the Technical Staff with Microelectronics and Computer Technology Corporation (MCC), Austin. Since 1994, he has been with the School of Electronic Engineering, Soongsil University, Seoul, where he is currently a Professor. From 2009 to 2010, he was a Visiting Professor with the Department of Electrical and Computer Engineering, The University of British Columbia, Vancouver, BC, Canada. He was the Dean of the School of Electronic Engineering, the Vice Dean of the College of IT, the Dean of the Office of Research and the Office of Planning, and the Provost and the Senior Vice President for Academic Affairs of Soongsil University. His research interests include mobile communications and intelligent signal processing.

Dr. Shin served as an Organizing/Technical Committee Member for various prominent international conferences, including IEEE VTC 2003-Spring, ISITA 2006, IEEE ISPLC 2008, APCC 2008, IEEE ISIT 2009, APWCS 2009, APWCS 2010, APCC 2010, APWCS 2011, ISAP 2011, APWCS 2012, APCC 2012, IEEE ICASSP 2018, and IEEE WCNC 2020. He was a recipient of the 2024 Journal of Communications and Networks Best Paper Award from KICS/IEEE ComSoc and the 2025 ICT Express Best Paper Award from KICS. In particular, he was the Technical Program Committee Chair of APWCS 2013, the General Co-Chair of APWCS 2014 and APWCS 2015, and the General Vice Chair of IEEE ICC 2022 and APCC 2022. He was the President of Korean Institute of Communications and Information Sciences (KICS) in 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:19:28 UTC from IEEE Xplore. Restrictions apply.